Safai Saathi ([redacted])

PS000051: Citizen-Centric Municipal & Rural Garbage Collection, Open Gutter & Public Sanitation Monitoring System

Panchmahal District Administration • Godhra Nagarpalika & Rural Gram Panchayats

OFFICIAL SYSTEM WIREFRAME & ARCHITECTURE SPECIFICATION

This specification outlines the end-to-end design system, screen wireframe blueprints, telemetry data flows, and accountability escalation matrices powering the Safai Saathi citizen mobile application.

1. System Topology & Ward Allocation

To ensure rapid, realistic ETAs (2–8 minutes) and eliminate driver fatigue, the 7 wards of Godhra and surrounding Panchayats are partitioned into three distinct vehicle clusters:

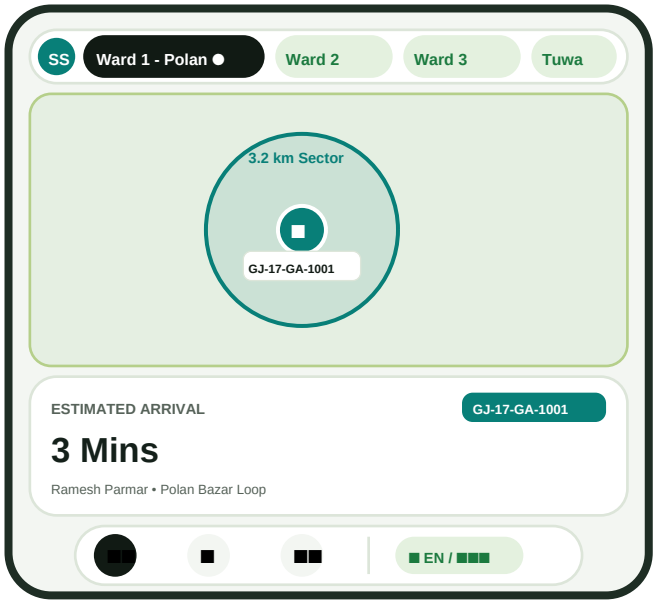
Ward Sector	Areas Covered	Vehicle & Model	Radius	Avg ETA
North Godhra (Commercial)	Ward 1 (Polan Bazar), Ward 2 (Station Rd), Ward 3 (Vavdi)	GJ-17-GA-1001 (Ashok Leyland Dost)	3.2 km	2 - 4 Mins
South Godhra (Residential)	Ward 4 (Bhuravav), Ward 5 (Jafrabad)	GJ-17-GA-1002 (Tata Ace Tipper)	3.5 km	3 - 6 Mins
Rural Panchayats	Tuwa & Bhamaiya Gram Panchayats	GJ-17-GA-1003 (Mahindra Bolero Maxi)	3.8 km	5 - 8 Mins

2. Design Tokens & Visual Hierarchy

Token	Hex Code	Functional Role
Deep Teal (Primary)	#087F78	Primary brand, app headers, active status indicators, official seal
Forest Green (Success)	#167B40 / #1B7A3F	Resolved states, collection complete cards, verified checkmarks
Civic Amber (Accent)	#F0873A	Alert chips, missed collection reports, map pin highlight
Crimson Red (Escalation)	#B3261E	72h SLA breaches, urgent red alerts, reopen status
Surface Neutral	#F3F6F2	Background canvas, pill chips, card containers

Screen 1: Live Map Telemetry & Dynamic Sector ETA

The primary dashboard of the Safai Saathi app provides real-time GPS telemetry of municipal solid waste collection vehicles, dynamic ETA calculations, and horizontal Ward sector selection.

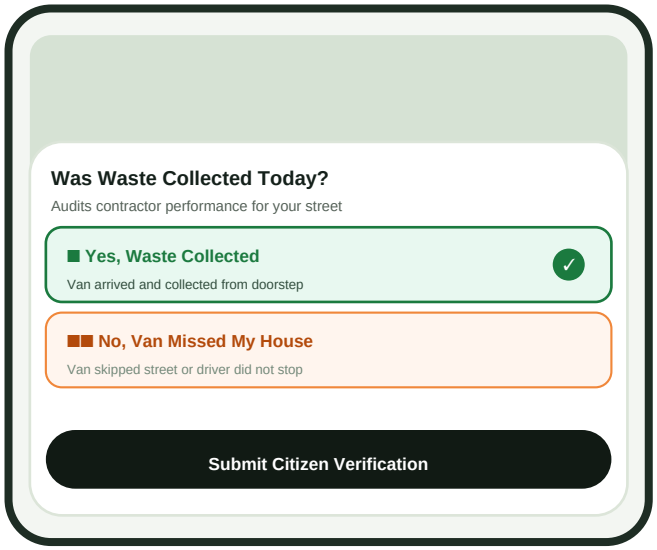


Key UX Specifications:

Component	UX Behavior & Technical Architecture
Ward Selector Carousel	Positioned at top-4 with horizontal momentum scrolling. Tapping any ward seamlessly centers the camera and draws the 3.0–3.8 km sector geofence.
Van Telemetry & Siren	Vehicles transmit coordinates via WebSocket/Polling. When within 500m of the citizen's location, the app plays an arrival audio alert.
Dynamic Bottom Sheet	Displays plate number, driver identity, and live ETA calculated via Haversine distance and real-time vehicle velocity.
Global Language Switcher	Relocated to the persistent bottom navigation bar. Accessible on every single screen without covering any content or buttons.

Screen 2: Doorstep Waste Collection & Citizen Verification

To prevent phantom municipal reporting, citizens verify whether garbage was collected from their premises. Submitting feedback dismisses the ETA card and updates the contractor performance score.



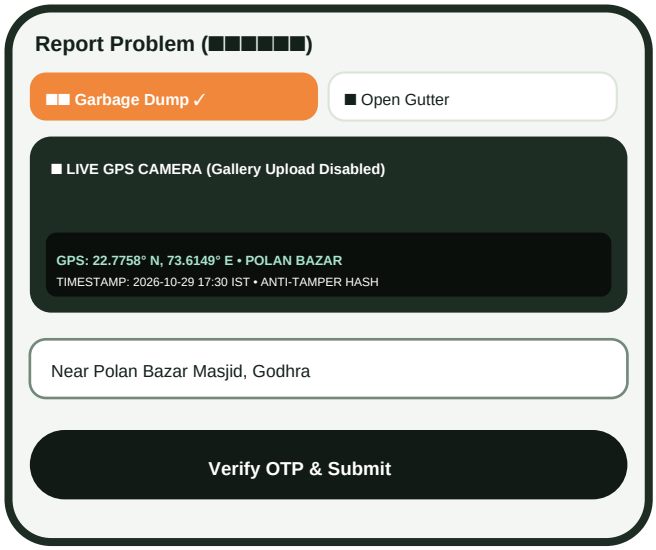
State Transition: ETA Disappearance Mechanism

Before Verification: The map bottom sheet displays the live ETA card (e.g. '3 Minutes').
After Marking 'Collected': The ETA card is permanently dismissed for the day and replaced by an emerald confirmation banner: ■ ■■■■ ■■■■ ■■■■■■■ ■■ ■■■■ ■■! (Today's Waste Collected).
Fail-Safe Reset: An 'Undo / Change Status' button is provided in case the citizen submitted by mistake.

Citizen Response	System Action	Contractor KPI Impact
Yes, Waste Collected	Dismisses ETA card, logs positive audit entry with GPS timestamp	+1.0 Completion Score (100% payout)
No, Van Missed House	Prompts reason (skipped lane / bin overflowed) and flags sector supervisor	-2.5 Penalty deduction if >5 reports in 300m
No Waste Generated	Logs premises as inactive/self-composting, dismisses ETA card	Neutral (excluded from missed quota)

Screen 3: Live Camera Grievance Redressal & OTP Verification

To eliminate fraudulent tickets and outdated gallery photos, citizens must capture live imagery through the in-app camera viewfinder. Every report requires fresh OTP verification without storing phone numbers.

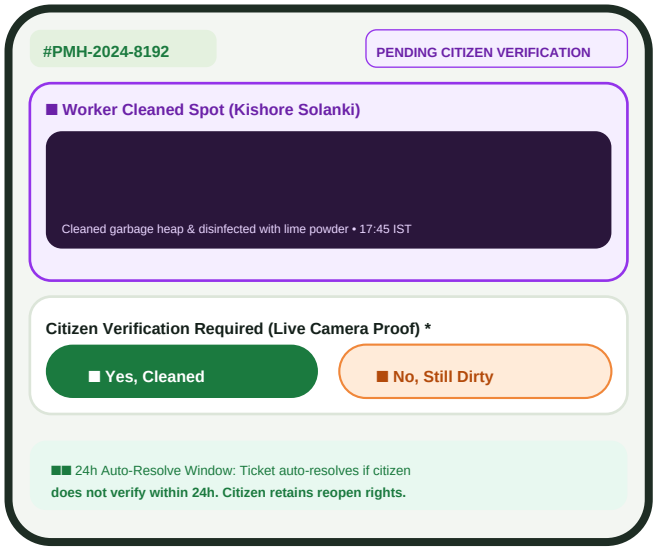


Cryptographic Proof & Privacy Guardrails:

Security Control	Implementation & Municipal Purpose
MediaStream Live Camera Only	HTML5 file input filepicker is disabled. Live camera canvas captures immediate ground reality, preventing reused stock images.
Embedded Geospatial Watermark	Latitude, longitude, GPS horizontal accuracy radius, and UTC timestamp are encoded onto the image buffer before submission.
Mandatory Fresh OTP	Citizens enter a 10-digit phone number verified via OTP (123456) on every report. Phone numbers are strictly never cached in localStorage.
In-App Official Ticket Receipt	Generates an official ticket (#PMH-2024-XXXX) assigned to the ward inspector with 24h SLA and direct navigation to 'My Complaints'.

Screen 4: Worker Cleanup & Citizen Cross-Verification Loop

In traditional portals, contractors mark issues resolved without proof. Safai Saathi enforces a mandatory two-way loop: workers upload cleaning photos, and citizens must cross-verify with camera evidence.

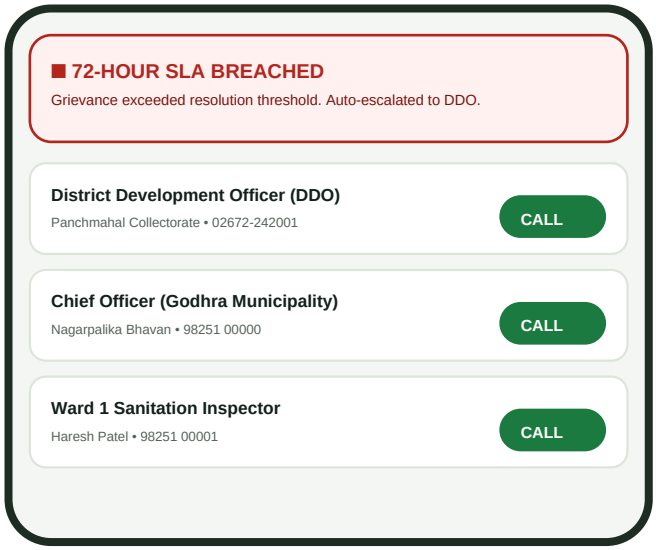


Cross-Verification Rules Matrix:

Step	Actor	Required Proof	Ticket Status
1. Cleanup Completed	Sanitation Crew	High-res after-cleaning photo with lime powder powdering	pending_verification
2A. Confirmed Clean	Reporting Citizen	Live camera snapshot verifying clean corner	resolved (Closed)
2B. Disputed (Dirty)	Reporting Citizen	Live camera snapshot of persisting filth + dispute reason	reopened (Escalates)
3. 24h Non-Response	System Automation	Auto-timer completes; audit log stamped	auto_resolved (Reopenable)

Screen 5: SLA Monitoring & 72h Auto-Escalation Engine

Grievances are bounded by legally enforceable municipal SLAs. If an issue remains unresolved past 24, 48, or 72 hours, it automatically escalates up the bureaucratic hierarchy to the Godhra Chief Officer and DDO Panchmahal.



SLA Multi-Tier Escalation Matrix:

Time Elapsed	Escalation Level	Designated Authority	Action Triggered
0 - 24 Hours	Level 0 (Standard)	Ward Sanitation Inspector	Work order assigned to localized sanitation crew
24 - 48 Hours	Level 1 Escalation	Chief Officer (Godhra Municipality)	Automated alert on CO dashboard; supervisor flagged
48 - 72 Hours	Level 2 Escalation	Executive Engineer (Sanitation)	Emergency crew dispatched; contractor penalty warning
> 72 Hours	Critical Breach	District Development Officer (DDO)	Collectorate oversight; contractor penalty applied; direct citizen hotline enabled